

Answers

- Q1.) (a) $2\sqrt{6}$ (b) $4\sqrt{5}$ (c) $5\sqrt{3}$
 (d) $2\sqrt{13}$ (e) $9\sqrt{2}$ (f) $6\sqrt{5}$

- Q2.) (a) $1008 = 2^4 \times 3^2 \times 7$
 (b) $12\sqrt{7}$

- Q3.) (a) $4\sqrt{2}$ (b) $3\sqrt{5}$ (c) $2\sqrt{6} - 17\sqrt{3}$

- Q4.) (a) 7 (b) 6 (c) 11 (d) 1275
 (e) $\sqrt{6}$ (f) $5\sqrt{2}$ (g) $7\sqrt{10}$ (h) 6
 (i) 150 (j) $44\sqrt{2}$ (k) $18\sqrt{10}$ (l) $15\sqrt{6}$

- Q5.) (a) $2 + 3\sqrt{2}$ (b) $5\sqrt{2} + 5$ (c) $3\sqrt{3} + 45$

- Q6.) (a) $1 + \sqrt{2} + \sqrt{3} + \sqrt{6}$ (b) $11 + 5\sqrt{5}$
 (c) $1 + \sqrt{5}$ (d) $59 + 14\sqrt{7}$
 (e) $75 + 16\sqrt{11}$ (f) $45 - 20\sqrt{5}$
 (g) 6 (h) 2
 (i) 1 (j) 24

- Q7.) Pythagoras' Theorem tells us that

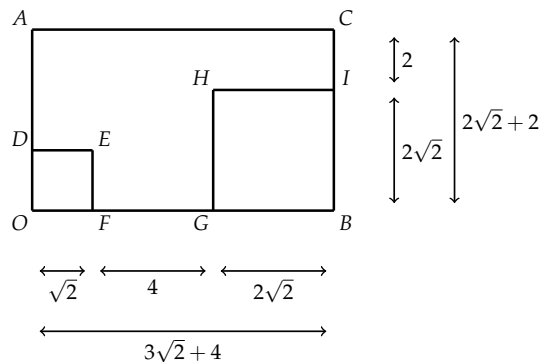
$$(3 + \sqrt{2})^2 + (3\sqrt{2} - 1)^2 = c^2 \quad (1)$$

After expanding brackets this becomes

$$(11 + 6\sqrt{2}) + (19 - 6\sqrt{2}) = c^2$$

Simplifying, we get $c^2 = 30$, so $c = \sqrt{30}$. Therefore the length of AB is $\sqrt{30}$.

- Q8.) Using the information, we can fill in the diagram. Since the area of $ODEF$ is 2, its sidelengths are $\sqrt{2}$. Since the area of $GHIB$ is 8, its sidelengths are $\sqrt{8} = 2\sqrt{2}$.



Using this, the area works out as

$$(3\sqrt{2} + 4)(2\sqrt{2} + 2)$$

Expanding this, we get a final answer of

$$(20 + 14\sqrt{2}) \text{ cm}^2$$

Q9.) (a) $\frac{\sqrt{10}}{10}$ (b) $\frac{5\sqrt{3}}{6}$ (c) $\frac{2\sqrt{7}}{21}$ (d) $\frac{\sqrt{15}}{5}$
 (e) $\frac{\sqrt{6} + 2\sqrt{3}}{6}$ (f) $\frac{2\sqrt{3} + 3}{21}$

Q10.) (a) $2 - \sqrt{3}$ (b) $\frac{30 + 5\sqrt{2}}{34}$ (c) $\frac{\sqrt{5} - 1}{4}$
 (d) $\frac{20\sqrt{3} + 6}{97}$ (e) $-\frac{5 + 3\sqrt{2}}{7}$ (f) $\frac{34 + 11\sqrt{7}}{3}$

Q11.) $3\sqrt{3} - 3\sqrt{2} + \sqrt{6} - 2$

Q12.) $(a + b\sqrt{2})^2 = 4\sqrt{2} + 6$. Expand the left-hand side: $a^2 + 2ab\sqrt{2} + 2b^2$.
 Since a and b are whole numbers, we can split this into two parts,

$$a^2 + 2ab\sqrt{2} + 2b^2 = (a^2 + 2b^2) + (2ab\sqrt{2})$$

and then match it up with $4\sqrt{2} + 6$ to get

$$4 = 2ab \quad (\text{From } \sqrt{2} \text{ coefficient})$$

$$\therefore 2 = ab$$

$$6 = a^2 + 2b^2 \quad (\text{From integer part})$$

The first equation tells us that a and b is a factor pair for 2, which there aren't many of, so we can use trial and error to get the answer of

$$a = 2, \quad b = 1$$

Note: It might look like $a = -2, b = -1$ works too, but this makes $\sqrt{4\sqrt{2} + 6}$ negative, which is not possible.